Appendices of Universe and Mind 6

Is the emergence emergence and emergent?, or Emergence is the evaluator-observer's confusion of the actual causality-control units causing the transformations; inappropriately matched scales and domains of the comparisons of incompatible properties and phenomena; inability to correctly segment, understand, predict and retrodict; and illegal and wishful hiding of the real underlying complexity and exchanging it with highly and lossy-compressed cherry-picked high-level representations and metrics which have lost their generative and transformational mapping to the real machine language of the lower-level universes

Todor Arnaudov ~1/2024[1]+

Appendix to "Universe and Mind 6" (UnM6) which is a volume and appendix to *"The Prophets of the Thinking Machines: Artificial General Intelligence and Transhumanism: History, Theory and Pioneers; Past, Present and Future"*

This article is inspired by discussions by M.Levin et al. about the experiment with evolving sorting algorithms of 6 lines of code[2].

How emergence is induced, perceived? States, results, data coming "unexpectedly" etc.

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[1] Written ~ 1.2024, a few short clarifications and edits on 21.9.2025; corrections: 28.3.2026. The title is a reference to "The Matrix in the Matrix is a matrix in the matrix", T.Arnaudov, 2003, "The Sacred Computer"
https://medium.com/@todorarnaudov/the-matrix-in-the-matrix-is-a-matrix-in-the-matrix-895e86c5f002

[2] Classical Sorting Algorithms as a Model of Morphogenesis: self-sorting arrays reveal unexpected competencies in a minimal model of basal intelligence, Taining Zhang, Adam Goldstein, Michael Levin
https://arxiv.org/abs/2401.05375
Discussions: e.g. * Karl Friston, Adam Goldstein, and Michael Levin discuss active inference and algorithms Michael Levin's Academic Content | 6,68 хил. Абонати | 3091 показвания 4.02.2024 г.
This is a 1-hour discussion meeting between Karl Friston, Adam Goldstein, and I talking about how Karl's active inference ideas apply to some work on unexpected behavior in sorting algorithms (https://thoughtforms.life/what-do-alg....
https://www.youtube.com/watch?v=ApVWEn-3vG4
*** K.F. 10 мин. Similarity of the alg. and of the content and the value... Markov random field

*** Comparing different domains, views and scales which are incompatible**. The classical example with fractals (in the video: *sorting algorithm*): the saying that *"how come so simple rules generate so complex phenomena"* etc.
1. The rules are simple only in one format of the representation **for a mind** with a particular sophistication in an abstract space – one complexity is traded off for another.
2. The *actual* rules for the universe require *complete* definition of the implementation.
3. Even in abstract spaces, the fractal is unfolded in *memory* and in *time* and these operations add up to the complexity, if they are considered as well.
4. The *surprise* from that *apparently new* complexity (unpredictability or a number of elements given particular segmentation/analysis, tokenization model etc.) comes from the evaluator-observer's *inability to predict or to see it*, without that *additional* memory and computation[3], with some desired or expected precision etc. – the *unpredictability* is part of the "computational irreducibility" from S.Wolfram's talks; thus complete "*science"*, or *a more compressed* representation and model, which could predict all details at the highest resolution of the output, given so short and *incomplete* definition as the one chosen for the abstract definition of the problem, may be impossible; in Theory of Universe and Mind (TOUM): the higher level/more abstract causality-control units predict with a lower resolution than the lower levels. The simple definitions are already compressed, they lose the detail of the originals. *However* as always, this depends on the observer's capabilities. One system might be incapable to predict or model an event in a "simple" way or with a target precision with *any* number of instructions or processing power, while for another system the same event or problem might be obvious and accessable as a one-step hash-map.
* If we evaluate even the most complex program, executed on a single CPU core, as a linear algorithm, as designed by the programmer, at each or a given "*shortest"* possible moment, it would be as complex as a single instruction or a span of the sequence of instructions, or the selected context with a given length (several instructions, a subroutine etc.) and the state of the registers of the processor, the memory, the I/O devices etc.; therefore *locally* and *momentary*, and given a selected, chosen *range* and *resolution*, *"the most complex program"* is not more complex than *the simplest program (or maybe "any program")* of length 1 or N instructions etc., or the difference is only directly reflecting the different types of possible instructions and the clocks they require etc. (MOV, CMP, JMP, DIV, …)[4]; the complexity is expanded and rendered through traversing all states and accumulating the differences. It is the same with the apparently simple algorithms for the observer. For a program, a sequence, a structure of any length or size, the observer-evaluator may know that the more complex program is more complex *only* if she can *remember the history/the whole range* and evaluate it altogether. Otherwise, all programs are at most as complex

[3] AFAIK M.Levin admits that a science is needed for that.
[4] A detailed review of this range would be complex. [21.9.2025]

as the maximum of the complexity per *"focus span"*, current range of evaluation etc.
* For a sample processor in such theoretical comparisons, it's more convenient and more explicit if simpler 8-bit or 16-bit CPUs are evaluated, in order to avoid or reduce the "spilling" of the execution stages and other complications etc.

5. The evaluator that estimates a higher complexity has a corresponding intelligence and a sufficiently large receptive/processing span and depth, which encompasses all range of details and finds "unexpected" or "beautiful" correlations (matches). Thus part of the complexity evaluation and appreciation comes *from the mind of the evaluator* and the complexity encoded *there*[5]. As in art, music, literature, science – one may have to be apt in the domain in order to understand and appreciate the technical mastery with a knowledge of the techniques (even though one can be *amazed* by being unable to imagine, predict etc. even if she doesn't understand).

6. → **Local unpredictability = global predictability ...**
As explained in the classical TOUM, as early as 2002, which was first derived from the random numbers – the laws of the random numbers suggest determinism, or as some call it "*superdeterminism*" – in order the probability distributions of the averages to work, all has to be synchronized and all events to counteract each other. C.Fields, and M.Levin in 2022: *"No finite agent can discover all of the causal influences that determine its own behavior, as global determinism logically disallows local determinism. Hence, global determinism assures "free will" from every (finite) local perspective [159].* [6]"

7. Are the *only-local* rules in cellular automata, cells etc. *actually/completely local*? Can they be? How locality is defined? Locality is a matter of degree, as are the scales/causality-control units in the segmentations, e.g. quarks, electrons, atoms, molecules, ... tissues, organs, systems, organisms, sets/systems of organisms ... also of humans with their hierarchies and interactions etc.. All parts are supposed to be in the

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[5] Compare J.Gorard on Entropy in the citations in the appendix *Listove of The Prophets of the Thinking Machines.*
[6] Competency in Navigating Arbitrary Spaces as an Invariant for Analyzing Cognition in Diverse Embodiments
by Chris Fields 1,†ORCID andMichael Levin 1,2,*, Entropy, 2022 https://www.mdpi.com/1099-4300/24/6/819
* The Free Will Theorem. Found. Conway, J.; Kochen, S., Phys. 2006, 36, 1441–1473.
* https://en.wikipedia.org/wiki/Hidden-variable_theory
https://en.wikipedia.org/wiki/Superdeterminism
See also:
* "Wolpert Theorem about mutual unpredictability and the impossiblity of subuniverses to predict with highest resolution of causality-control are rediscoveries of concepts from Theory of Universe and Mind – entangled with unnecessary mathematical notations and unsatisfiable premises", chapter from "Listove" or "Reflections on Everything", the second biggest volume from "The Prophets of the Thinking Machines", T.A.2025

same Universe or the same environment, "ecological niche", body/organism, organ, tissue etc. They are all local and global in the same time, depending on the *evaluator's choice of resolution* and selection of parameters, boundary conditions, simulation/comparison precision etc. The local rules work for the whole system, because there are higher order ones, which monitor and supervise them (or there is an observer-evaluator, who *sees* such rules or interprets the transformations of the state of the system as “rules” which are *localized* to this-or-that causality-controlu unit, “locality”, space, subspace etc.). ...

What exactly counts as "*explicit*" rules/governing and why? One way for dealing with this segmentation/definition and for other tentative choices is the following: the choices which are already made “cover”/check/fill slots of the possibilities and that way they *disallow* some other overlaps and interpretations etc. The earlier selected rules limit and “lock” the others etc. and step-by-step this process constructs more or less exhaustive ontology, graph, structure etc., which covers the possibilities in a given universe/space/domain etc.

Qualitatively though, as of *having or not having an impact*, time and space shouldn't matter: the parts are connected with different delays, *time lags*. The independence of an atom, cell, tissue is up to some range and resolution. Sooner or later each of these elements or entities will interact with others, or it *always* does, given a resolution of causality control and perception, superimposing many influences – some of the latter are below particular threshold and are smoothed and overlapped by other impacts which are stronger and counteract or mask the small impacts and influences[7] – or they are masked or *appear* as smoothed or *masked for an evaluator-observer with too low a resolution to detect them*, because of a limited span which is required to notice in space and time earlier or more distant influences; with limited capabilities to see in the past and deduce the origins: to “retro-predict” - to rewind the simulation back a required number of transformations and interactions in order to detect some of the influences etc.

Similarly, FEP/AIF (Free Energy Principle/Active Inference) assumes the Markov blankets are conditionally *independent* and the communcation happens *only* through the boundary/perception and the *”action states”*, but it's all within a given resolution and error and that definition is "in abstracto". If the universe at some current/observed level is perceived or modeled, or if it *can be* perceived /modeled /represented as built by uniform or just by *lower level building blocks*, which are

[7] I think this was noted somewhere in Levin's or Friston's literature regarding the separation of the scales/levels, it is related/could be connected with the *precision* of the predictions, generative models – each scale/level, informationally, as of numerical value of sensory-motor output, could be defined as of the precision of the prediction/causation, represented as numbers. If it's 0,56 with a precision/resolution of +-0,01, it could be between 0,55 or 0,57, or given some distribution: with a probability etc. In more abstract domains it could be an item/class from a set, say an animal (dog, horse, cat, elephant, ...)

interacting, then at the lower *representation* level, or "mother universe", or "physics", or "machine language of the universe", from the POV of the current virtual universe (current resolution of representation of an agent, current "Markov blanket"), *that conditional independence is lacking*. All structures, defined at higher compositional levels, are generated by the *common* generative process of the Universe and the abstractions are “leaking”. [11-1-2024+ editions]

* A related case is Levin's work on the *sorting algorithms*, local rules etc. Six lines of code are exercised in order to demonstrate that a piece of code that is so simple can produce so complex behavior etc.

Here again there is **a mixing of different contexts and scales**. The code as a record of 6 lines is *static*, it's encoded in a high level abstract language which requires humans or/and computers in order to be translated into the machine language of some actual "living" virtual universe. However, neither the *interpretation*, which converts that 6-lines to the language of the physical universe is just 6 lines of *that* simple code, nor the encoding of the representation of the evalutor-observer is so short in *its/her machine code*. Furthermore the **executed** code, even in a computer, is *way more* than the initial code, even if it is programmed in machine code at “bare metal” level for EDSAC: it *resides in memory*, which is supposed to be bigger and the *being* in that memory introduces and implies other, new, additional rules, information, context, possibilities etc. When the program is "executed", it becomes *causal* and connected to the fabric of the substrate where it runs, and finally it expands in time, so its area, volume, the memory that it covers in the univese is growing. *The last item*, linked with the limited working memory/span of the evaluator-observers – what one can understand, see etc. at once, – and *the desire to be amazed* by selecting the right properties – makes the observed patterns appear "unexpected", "emergent", "unpredictable", "magical".

The inability to predict though is a sign namely of *inferior predictive capacity than required for the problem or for understanding it* – the capability for prediction is a measure of general intelligence by the definitions in TOUM, be it in general or in the particular domain or case, where the intelligence may fail short due to a lack of sufficient experience, appropriate records and eventually after collecting them, a given phenomenon could become "sufficiently" predictable, so that it is not "surprising" anymore. ...

Perplexity, Kullback-Liebler divergence ... See *Universe and Mind 3*, #10. In the framework of TOUM and the basic definitions of the causality-control units, if the evaluator-controller cannot cause, control, write to the memory of the phenomenon with the maximum resolution of causality and perception in the target / supposedly subordinate / controlled universe, then its causality-control is only *virtual*, up to some lower resolution, i.e. the prediction errors will accumulate and grow, and the **actual*, **real* causality control unit which can predict *exactly wih no errors* is *something else*,

elsewhere; it is another causality-control unit.

In the context of the 6-lines-of-code – these lines do not include *the definitions of the system that executes the code*, the memory, the representation and the states, extended in time etc. – these other requiremetnt are implied and forgotten in the calculations.

The actual causality control unit *is not *only* this code* and the abstractions are not enough to describe the machine code of actual entities in a physical universe. One may object that not only the 6-line program, but more complex initial programs also have the underlying executing machine etc. as a compagnion, which is true, however this remark reminds also that *the bulk of the complexity is not in the definition of this program code*, but namely *in the underlying "physics"* of the lower-level representation of the Universe. One algorithm is 6 lines of code, another is 1000 and the third is 1 million. How many and what kind of machines or mechanisms are needed in order to define the explicit state and conditions at the resolution of the known physical particles even just of the computer that runs this code, defined as a *physical object*, even if it is an 8-bit microcomputer with 6502 CPU – say an Apple][?

CPUs and more sophisticated minds/predictors:

* Simple CPUs have simple basic instructions and need many of them and more complex logic for a given task, e.g. the 8-bit 6502, the CPU of Apple][, Commodore 64 etc. early PCs; Atari 2600, NES etc. video game consoles etc. has only one 8-bit accumulator and two 8-bit index registers. There is no instruction for multiplication and no 16-bit registers, which makes programming the microprocessor more difficult in comparison to more advanced 8-bit CPUs – for example Z80. The Z80 has 16-bit registers and more complex instructions and can do in one instruction the same job that requires complex subroutines in 6502. Even more advanced and higher word size CPUs encompass more and more functions in single instructions, they introduce branch prediction, out-of-order and speculative (predictive) execution[8]. A future that is unpredictable even just one instruction ahead from the "blade" of the "focus" of the program counter of a 6502 for hundreds or thousands of instructions ahead, for example if it has to execute the unfolding of a single AVX SIMD etc. instructions with floating point operations, is executed in just one instruction of 2010 x86 CPU, in one of its many hyperthreading execution units, in one of its subunits. The informational, temporal and spatial *future* of a 6502 for a millisecond or a second can be encoded in the crystal of the *future* CPU as a "web" of activations and states of the transistors within a *nanosecond*. The equivalent 6502 program is spread in the silicon and the external program is just one instruction – this complexity is traded off for the

[8] In the microprocessors - Pentium I introduces branch prediction, released in 1993; out-of-order execution comes in the next generation Pentium Pro. The mainframe IBM S/360 91 had out-of-order execution in 1966.

technological advancements and all R&D and scientific operations and the Universe operations required in order to produce the future CPU, to put it into the socket, to develop the computer science and the software etc.

* The complexity and “optimization” trade-offs – there is no real optimization – are studied more deeply in the appendix to “The Prophets…”:
“Is Mortal Computation Required for the Creation of Universal Thinking Machines?”, T.Arnaudov, SIGI-2025
It was published in Bulgarian, original title:
* Нужни ли са смъртни изчислителни системи за създаване на универсални мислещи машини?, Т.Арнаудов, SIGI-2025
https://github.com/twenkid/sigi-2025

Notes … Continue, refine:
-- More advanced CPUs …
One automatic assumption is regarding time, things which happen "in the same time". ... Also, qualitatively the time may not …
-- Solving the same problem with different means (M.L., W.James)
Is it then *the same problem actually,* though? The criterion?
Last corrections: 28.3.2026